Combinatorics of Large Covers

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Context and Background

Throughout, for all topological spaces considered, we assume NO separation axioms, and will consequently be explicit when any are used.

(Though nothing would be lost by taking Kolmogorov quotients and assuming all spaces are at least T_0.)

Cover types

- For a topological space X, let $\mathcal{O}_X$ denote the set of all nontrivial open covers of X; an open cover $\mathscr{U}$ is *nontrivial* if $X \notin \mathscr{U}$.
- A cover $\mathscr{U} \in \mathcal{O}_X$ is a *large* cover if, for every $x \in X$, $\{U \in \mathscr{U} : x \in U\}$ is infinite. We let Λ_X denote the collection of all large covers of X.
- A cover $\mathscr{U} \in \mathcal{O}_X$ is an *ω-cover* if, for every finite $F \subseteq X$, there is some $U \in \mathscr{U}$ such that $F \subseteq U$. We let Ω_X denote the collection of all ω-covers of X.
- A cover $\mathscr{U} \in \mathcal{O}_X$ is a *k-cover* if, for every compact $K \subseteq X$, there is some $U \in \mathscr{U}$ such that $K \subseteq U$. We let $\mathcal{K}_X$ denote the collection of all k-covers of X.

Three Trivial Observations

Recall the equivalence relation on a nonempty space X where $x \sim y$ if the neighborhood filters of x and y are identical. Then the Kolmogorov quotient is simply $X/\sim$.

A space is said to be *topologically finite* if its Kolmogorov quotient is finite.

- For any space X, $\Omega_X = \varnothing$ if and only if X is topologically finite.
- For any space X, $\mathcal{K}_X = \varnothing$ if and only if X is compact.
- For any space X, $\Lambda_X = \varnothing$ if and only if some point has a finite open neighborhood filter.

As a consequence of the nontriviality condition on the covers, we have that

$$\mathcal{K}_X \subseteq \Omega_X \subseteq \Lambda_X \subseteq \mathcal{O}_X.$$

To see that $\Omega_X \subseteq \Lambda_X$, let $\mathscr{U} \in \Omega_X$ and $x \in X$ be arbitrary.

- For $n \in \omega$, suppose $F_n := \{x\} \cup \{x_j : j < n\} \in [X]^{n+1}$ has been defined.
- Let $U_n \in \mathscr{U}$ be such that $F_n \subseteq U_n$.
- $U_n \neq X$, so there is $x_{n+1} \in X$ such that $x_{n+1} \notin U_n$ (and so $x_{n+1} \notin F_n$).
- $x \in U_n$ for every $n \in \omega$ and $\{U_n : n \in \omega\}$ is infinite as it consists of pairwise distinct elements.

Selection Principles

Let $\mathcal{A}$ and $\mathcal{B}$ be sets. Then

$$\mathsf{S}_1(\mathcal{A},\mathcal{B}) \equiv (\forall A \in \mathcal{A}^\omega)\left(\exists B \in \prod_{n\in\omega} A_n\right)\ \{B_n : n \in \omega\} \in \mathcal{B}$$

and

$$\mathsf{S}_{\mathrm{fin}}(\mathcal{A},\mathcal{B}) \equiv (\forall A \in \mathcal{A}^\omega)\left(\exists B \in \prod_{n\in\omega} [A_n]^{<\omega}_{+}\right)\ \bigcup\{B_n : n \in \omega\} \in \mathcal{B}.$$

Following Scheepers, $X \models \mathsf{S}_*(\mathcal{A},\mathcal{B})$, where $* \in \{1, \mathrm{fin}\}$, means that X satisfies the selection principle $\mathsf{S}_*(\mathcal{A}_X,\mathcal{B}_X)$.

E.g., $X \models \mathsf{S}_1(\mathcal{O},\mathcal{O})$ means that X is Rothberger.

Selection Games

Each of the selection principles above have corresponding games for two players, P1 and P2:

- $\mathsf{G}_1(\mathcal{A},\mathcal{B})$ and
- $\mathsf{G}_{\mathrm{fin}}(\mathcal{A},\mathcal{B})$.

In round $n \in \omega$, P1 chooses $A_n \in \mathcal{A}$ and P2 makes a single or nonempty finite selection from A_n, depending on the game. P2's goal is to create a member of $\mathcal{B}$ with their selections, and P1 wins the game if P2 fails to do so.

$*$	fin	1
$\mathsf{G}_*(\mathcal{O},\mathcal{O})$	Menger game	Rothberger game
$\mathsf{G}_*(\Omega,\Omega)$	ω-Menger game	ω-Rothberger game
$\mathsf{G}_*(\Lambda,\Lambda)$	λ-Menger game	λ-Rothberger game
$\mathsf{G}_*(\mathcal{K},\mathcal{K})$	k-Menger game	k-Rothberger game

Perfect-information Strategies

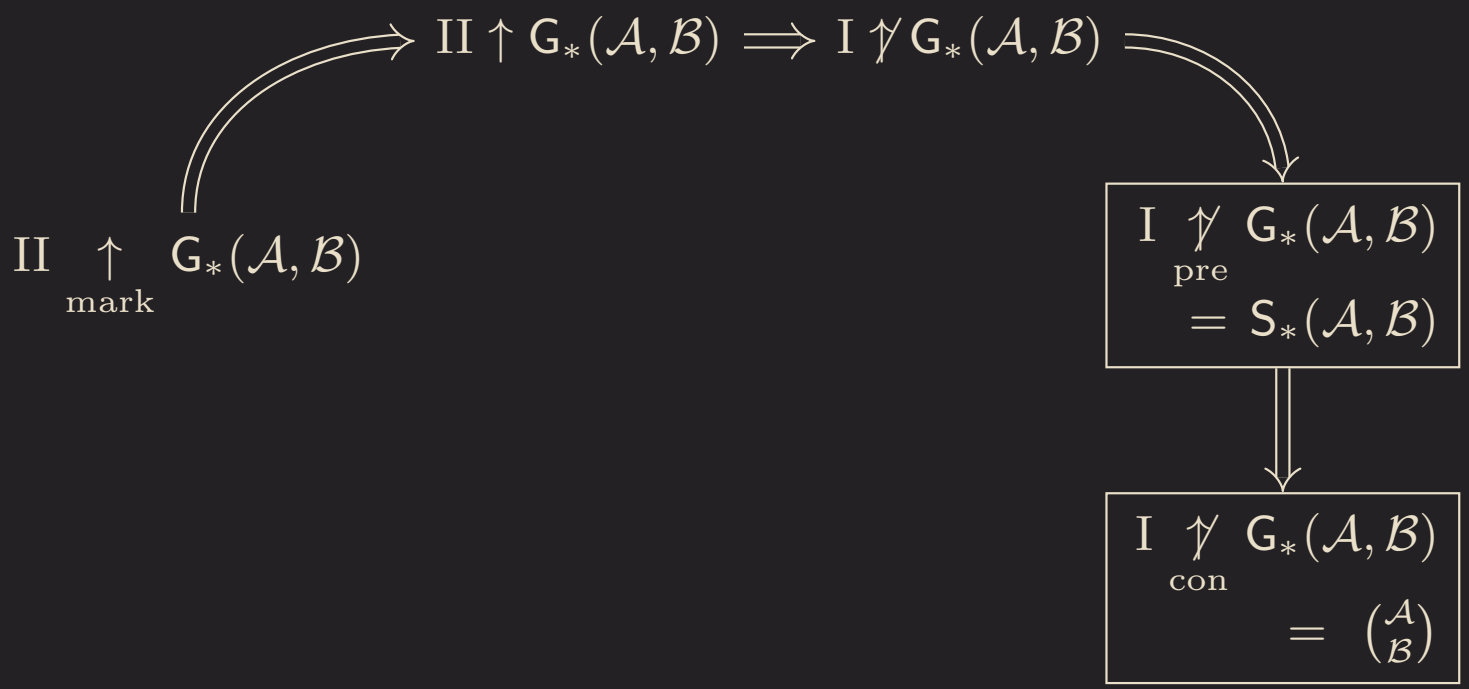
A winning perfect-information strategy for either player is a function which has as its domain all valid initial segments of play and outputs a valid play in each round such that the corresponding player wins any run of the game.

Limited-information Strategies

Limited-information strategies are strategies with nontrivially limited domains. Here, we will discuss primarily

- *constant* strategies for P1, which correspond to a single selection from $\mathcal{A}$,
- *predetermined* strategies for P1, which correspond to $\sigma : \omega \to \mathcal{A}$, and
- *Markov* strategies for P2, which correspond to functions $\tau : \mathcal{A} \times \omega \to \bigcup \mathcal{A}$ or $\tau : \mathcal{A} \times \omega \to [\bigcup \mathcal{A}]_{+}^{<\aleph_0}$, depending on the game.

We first note a straightforward flow of strategic strength:



Though the implications do not reverse, in general, an important component to selection game theory is to determine when selection principles are equivalent to P1 not having a perfect-information winning strategy.

- (Hurewicz, 1925)

$$\mathrm{I} \uparrow \mathsf{G}_{\mathrm{fin}}(\mathcal{O}_X, \mathcal{O}_X) \iff \mathrm{I} \underset{\mathrm{pre}}{\uparrow} \mathsf{G}_{\mathrm{fin}}(\mathcal{O}_X, \mathcal{O}_X)$$

- (Pawlikowski, 1994)

$$\mathrm{I} \uparrow \mathsf{G}_1(\mathcal{O}_X, \mathcal{O}_X) \iff \mathrm{I} \underset{\mathrm{pre}}{\uparrow} \mathsf{G}_1(\mathcal{O}_X, \mathcal{O}_X)$$

- (Scheepers, 1997)

$$\mathrm{I} \uparrow \mathsf{G}_*(\Omega_X, \Omega_X) \iff \mathrm{I} \underset{\mathrm{pre}}{\uparrow} \mathsf{G}_*(\Omega_X, \Omega_X)$$

- (Scheepers, 1996/1999)

$$\mathrm{I} \uparrow \mathsf{G}_*(\Lambda_X, \Lambda_X) \iff \mathrm{I} \underset{\mathrm{pre}}{\uparrow} \mathsf{G}_*(\Lambda_X, \Lambda_X)$$

- (C. & Holshouser, 2022)

$$\mathrm{I} \uparrow \mathsf{G}_*(\mathcal{K}_X, \mathcal{K}_X) \iff \mathrm{I} \underset{\mathrm{pre}}{\uparrow} \mathsf{G}_*(\mathcal{K}_X, \mathcal{K}_X)$$

With

- the equivalences above and
- the observation that the constant strategy level for P1 is a form of Lindelöf-like property,

we will focus on the following three strategic layers:

P2 Markov, P2 Strategic, and P1 Strategic.

The P1 side of things has been thoroughly studied and most of the P1-related results here are due to Scheepers.

Why care about P2?

One reason:

- Selection principles are not, in general, productive.
- The P2-strategic analog is, by contrast, typically productive.

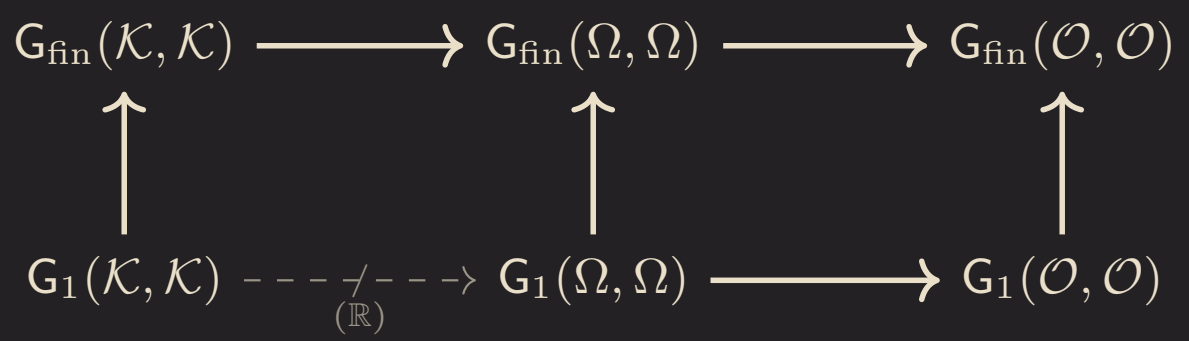
Another common aspect of selection game theory is to compare various games, either through game equivalence or game duality. To expand on the classical notion of game equivalence, which is defined only in terms of perfect-information strategies, we include limited-information strategies in the following way:

$\mathsf{G}_*(\mathcal{A},\mathcal{C}) \leqslant^+_{\mathrm{II}} \mathsf{G}_*(\mathcal{B},\mathcal{D})$

- $\mathrm{II} \underset{\text{mark}}{\uparrow} \mathsf{G}_*(\mathcal{A},\mathcal{C}) \implies \mathrm{II} \underset{\text{mark}}{\uparrow} \mathsf{G}_*(\mathcal{B},\mathcal{D})$
- $\mathrm{II} \uparrow \mathsf{G}_*(\mathcal{A},\mathcal{C}) \implies \mathrm{II} \uparrow \mathsf{G}_*(\mathcal{B},\mathcal{D})$
- $\mathrm{I} \not\uparrow \mathsf{G}_*(\mathcal{A},\mathcal{C}) \implies \mathrm{I} \not\uparrow \mathsf{G}_*(\mathcal{B},\mathcal{D})$
- $\mathrm{I} \underset{\text{pre}}{\not\uparrow} \mathsf{G}_*(\mathcal{A},\mathcal{C}) \implies \mathrm{I} \underset{\text{pre}}{\not\uparrow} \mathsf{G}_*(\mathcal{B},\mathcal{D})$
- $\mathrm{I} \underset{\text{con}}{\not\uparrow} \mathsf{G}_*(\mathcal{A},\mathcal{C}) \implies \mathrm{I} \underset{\text{con}}{\not\uparrow} \mathsf{G}_*(\mathcal{B},\mathcal{D})$

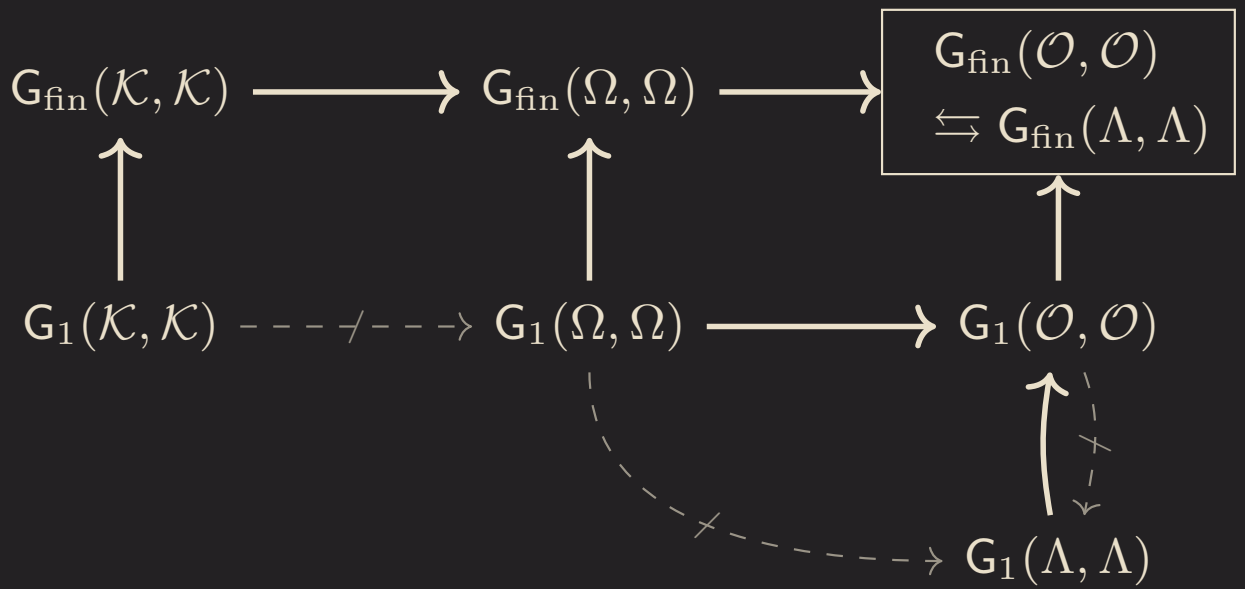
$\mathsf{G}_*(\mathcal{A},\mathcal{C}) \leftrightarrows \mathsf{G}_*(\mathcal{B},\mathcal{D})$

$\mathsf{G}_*(\mathcal{A},\mathcal{C}) \leqslant^+_{\mathrm{II}} \mathsf{G}_*(\mathcal{B},\mathcal{D})$ and $\mathsf{G}_*(\mathcal{B},\mathcal{D}) \leqslant^+_{\mathrm{II}} \mathsf{G}_*(\mathcal{A},\mathcal{C})$



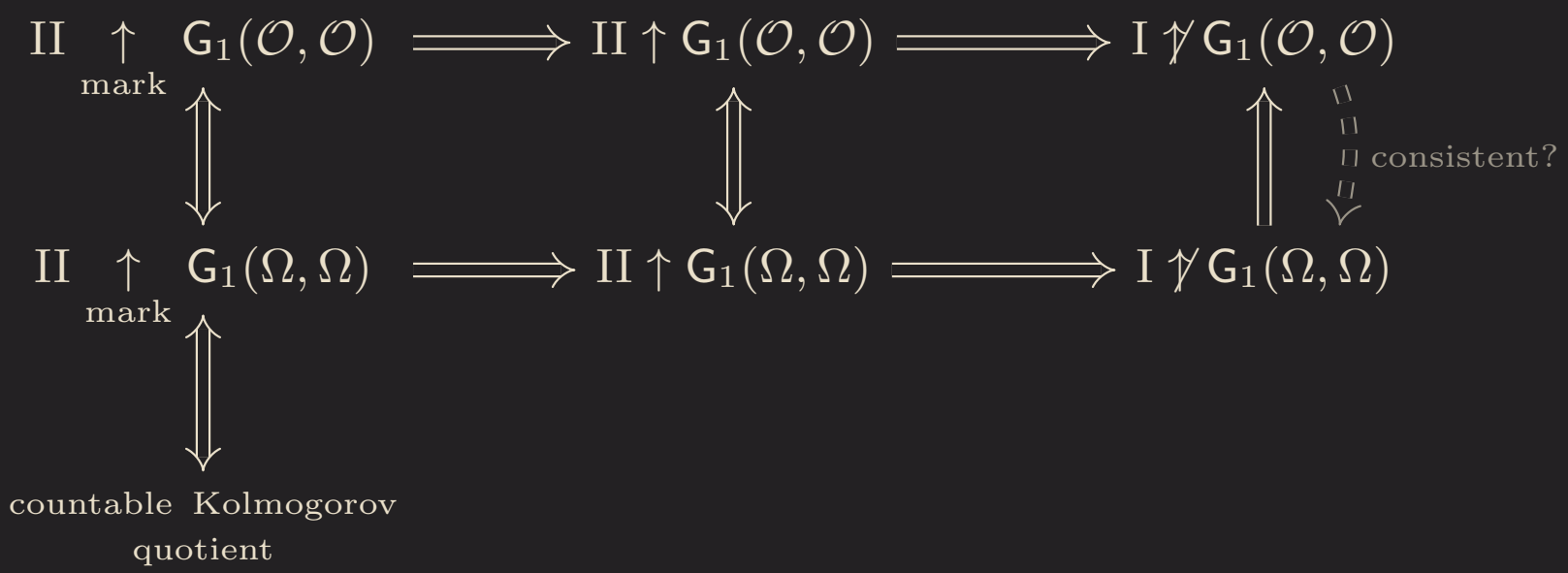
- P1 side: Scheepers diagram and Kočinac's work on k-covers.
- P2 side: completed in C., Clontz, & Holshouser, 2024.

Today's Main Result

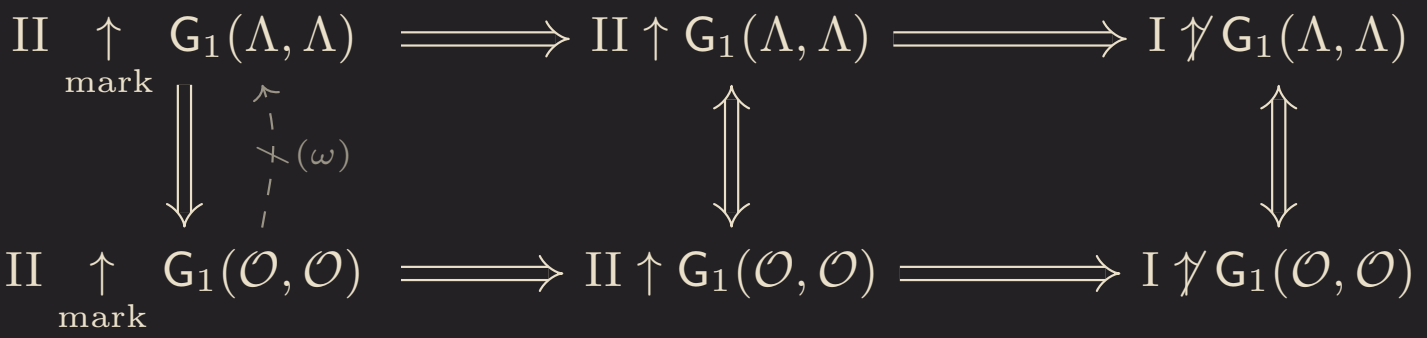


- P1 side: Scheepers + Kočinac.
- P2 side: completed in C. & Holshouser, 2026.

Recap on ω-Rothberger v. Rothberger



Placing $\mathsf{G}_1(\Lambda, \Lambda)$



Key ingredients for the affirmatives:

- On the P1 side: Pawlikowski + Scheepers
- For P2 perfect-information: P2 can pretend like she's playing infinitely many copies of the game while keeping track of her tail and making sure she never makes repeated choices.

In the Markov setting, P2's forgetfulness comes with a price.

Theorem (C. & Holshouser, 2025)

If X is a countably infinite T_1-space with an uncountable topology, then X is NOT Markov λ-Rothberger.

Corollary

No countably infinite Hausdorff space is Markov λ-Rothberger.

Proof of the corollary.

- Any infinite Hausdorff space admits a family $\{V_n : n \in \omega\}$ of nonempty, pairwise disjoint open sets.
- The mapping $\langle b_n : n \in \omega \rangle \mapsto \bigcup\{V_n : b_n = 1\}$ from 2^ω to the topology of X is an injection.

□

Proof of the theorem.

- Assume X is countably infinite with an uncountable topology. Then there must be some point $p \in X$ such that the open neighborhood filter $\mathcal{N}_p$ at p is uncountable.
- Let $\tau : \Lambda_X \times \omega \to \mathscr{T}_X$ be any function with the property that $\tau(\mathscr{U}, n) \in \mathscr{U}$.
- For $n \in \omega$, let $A_n = \bigcap\{\tau(\mathscr{U}, n) : \mathscr{U} \in \Lambda_X\}$.
- If $M_p = \{n \in \omega : p \in A_n\}$ is finite, for each $n \in \omega \setminus M_p$, choose $\mathscr{U}_n \in \Lambda_X$ such that $p \notin \tau(\mathscr{U}_n, n)$. For arbitrarily chosen $\mathscr{U}_n \in \Lambda_X$ when $n \in M_p$, note that $\{n \in \omega : p \in \tau(\mathscr{U}_n, n)\} \subseteq M_p$, and so τ is not winning for P2.

Proof (continued).

- Otherwise, $M_p = \{n \in \omega : p \in A_n\}$ is infinite.
- For each $n \in M_p$, let $\mathcal{E}_n = \{\tau(\mathscr{U}, n) : \mathscr{U} \in \Lambda_X\}$. ($p \in \bigcap \mathcal{E}_n = A_n$.)
- *Claim:* $\mathcal{E}_n$ is cofinite in $\mathcal{N}_p$.
  - For $x \in X \setminus \{p\}$, let

$$\mathcal{F}_x = \{X \setminus \{p, y\} : y \in X \setminus \{x\}\}$$

    and

$$\mathscr{U} = (\mathcal{N}_p \setminus \mathcal{E}_n) \cup \bigcup \{\mathcal{F}_x : x \in X \setminus \{p\}\}.$$

  - If $\mathcal{N}_p \setminus \mathcal{E}_n$ were infinite, $\mathscr{U}$ would be a large cover and, since $n \in M_p$, $\tau(\mathscr{U}, n) \in \mathcal{E}_n \not\subseteq \mathscr{U}$. $\rightarrow\leftarrow$

Proof (continued).

- $\bigcup\{\mathcal{N}_p \setminus \mathcal{E}_n : n \in M_p\}$ is countable and $\mathcal{N}_p$ is uncountable. So we can choose

$$U_p \in \mathcal{N}_p \setminus \bigcup\{\mathcal{N}_p \setminus \mathcal{E}_n : n \in M_p\} = \bigcap\{\mathcal{E}_n : n \in M_p\}.$$

- For each $n \in M_p$, choose $\mathscr{U}_n \in \Lambda_X$ such that $U_p = \tau(\mathscr{U}_n, n)$.
- For each $n \in \omega \setminus M_p$, choose $\mathscr{U}_n \in \Lambda_X$ such that $p \notin \tau(\mathscr{U}_n, n)$.
- Then $\{\tau(\mathscr{U}_n, n) : n \in \omega\} \notin \Lambda_X$, so P2 has lost.

□

Proposition

If X has a countable topology and a countably infinite Kolmogorov quotient, then X is Markov λ-Rothberger.

Proof.

- Faithfully enumerate the topologically distinguishable points of X as $\{x_n : n \in \omega\}$.
- Faithfully enumerate the topology on X as $\{U_n : n \in \omega\}$ and let $n \overset{\varphi}{\mapsto} U_n$.
- Fix a bijection $\beta : \omega^2 \to \omega$ which is increasing in the second coordinate; i.e., $n < m \implies \beta(t, n) < \beta(t, m)$, $t \in \omega$.
- So, define $\tau : \Lambda_X \times \omega \to \mathscr{T}_X$ to be a choice $\tau(\mathscr{U}, n) \in \mathscr{U}$ such that, letting $(a, b) = \beta^{-1}(n)$, $x_a \in \tau(\mathscr{U}, n)$ and $\varphi^{-1}\tau(\mathscr{U}, n) \geqslant b$.

□

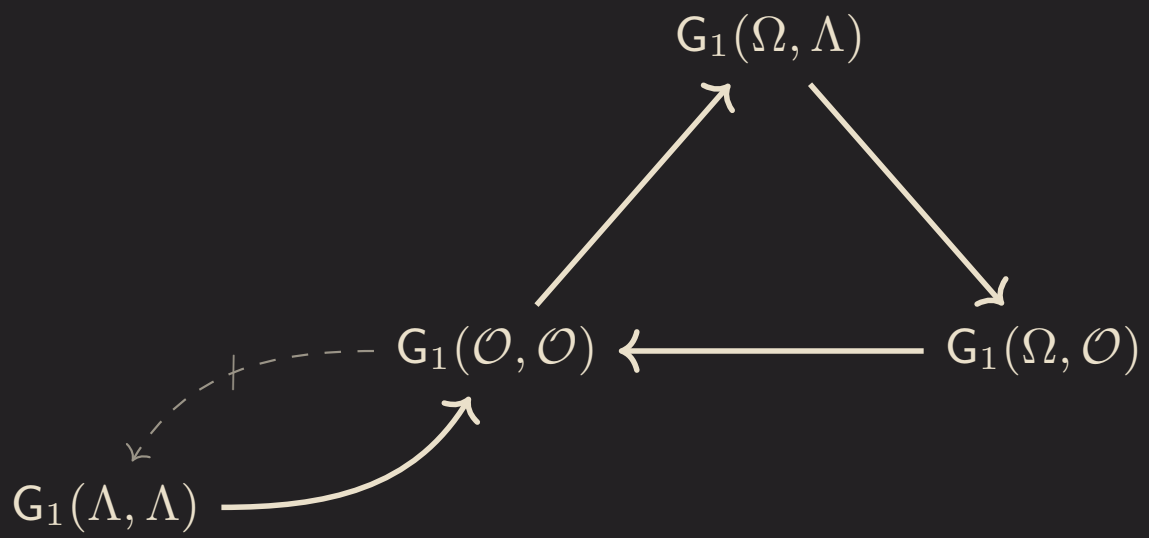
Corollary

If X is a T_1-space, then X is Markov λ-Rothberger if and only if X has a countable topology.

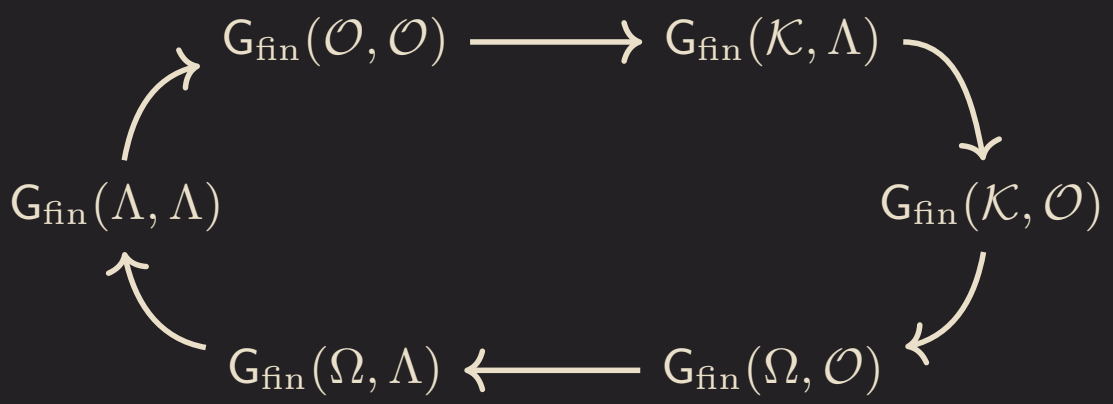
Question

Can the T_1 condition be dropped, or is there an example of a Markov λ-Rothberger space with an uncountable topology and a countable Kolmogorov quotient?

Summary of single-selection equivalences



Summary of finite-selection equivalences



Key ingredients:

- Closing any cover under finite unions produces either a k-cover or a trivial cover, and a finite selection of finite subsets of a collection corresponds to a finite selection from the collection.
- On the P1 side: Hurewicz + Scheepers
- For P2 perfect-information: P2 can pretend like she's playing infinitely many copies of the game while keeping track of her tail and making sure she never makes repeated choices.
- For the Markov layer: P2 can make larger and larger finite selections to guarantee new open sets have been chosen in each round.

Thank You!